

Process Implementation And Automation

Lecture 08

BUSINESS PROCESS MANAGEMENT
IS 2206

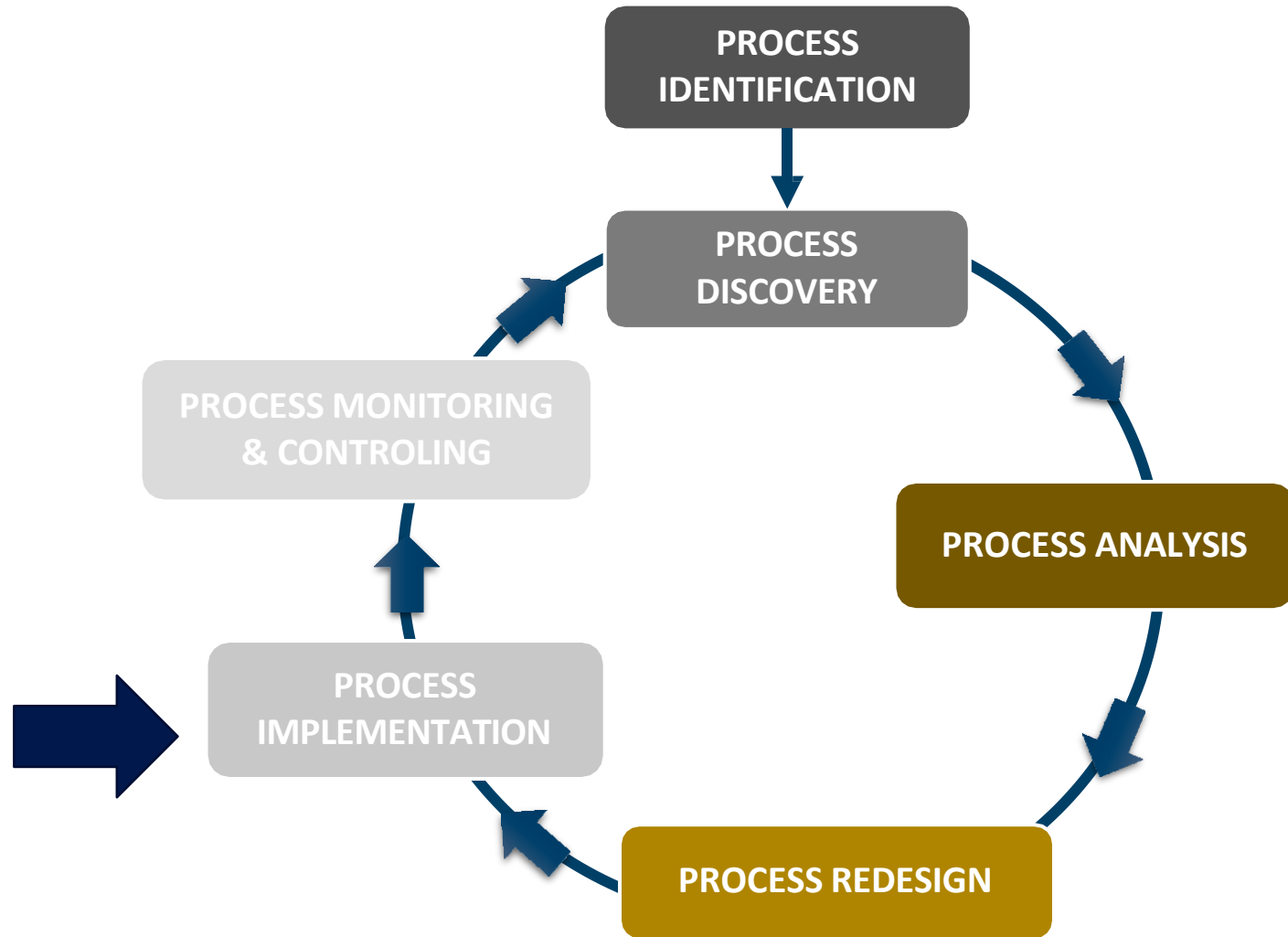
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OUTLINE

- What is Process automation?
- Process Aware Information Systems
- Business Process Management Systems
 - Architecture of a BPMS
 - Advantages of introducing a BPMS
 - Challenges in introducing a BPMS
- Turning Process Models Executable



WHERE
ARE WE
NOW?



TOPIC 01

What is Process Automation



1.1 What?

The approach of **automating** any **conceivable part** of procedural work that is contained within a business process, from simple operations that are part of a **single process** activity up to the automated coordination of entire, complex processes



1.1 What?

- Process automation means using technology to perform part or all of a business process automatically.
- Automation can be applied to very small tasks, such as sending an email, or to an entire business process involving many different activities.
- The main goal is to reduce manual work while improving speed and accuracy.
- **Example**
 - ✓ Instead of an employee manually sending confirmation emails to customers, the system automatically sends the email immediately after an order is placed.

1.2 What is an automated business process?

An automated business process refers to a sequence of **tasks, activities, or operations** within an organization that are executed using **automated technology** and **systems**, often without the need for **manual intervention**.

Automation involves using **software, tools, and technologies** to streamline and execute tasks, reducing the reliance on human involvement for routine and repetitive processes.



TOPIC 02

Process Aware Information Systems



2.1 What?

- Process-Aware Information System (PAIS) is a software system that **manages** and **executes** operational processes involving **people**, **applications**, and/or **information sources** on the basis of process model.
- The system knows what activity should happen next and who should perform it.
- This ensures that every process follows the designed workflow."
- **Example**
 - ✓ A university student registration system guides students, lecturers, and administrators through each step in the correct order.



2.2 Types of Process Aware Information Systems

Business Process Management (BPM) :

Business Process Management (BPM) systems are software platforms designed to model, automate, monitor, and optimize various business processes within an organization.

Customer Relationship Management Systems (CRM):

Customer Relationship Management (CRM) systems are software solutions designed to manage interactions and relationships with customers.

Enterprise Resource Planning Systems (ERP):

Enterprise Resource Planning (ERP) systems are integrated software solutions designed to manage and automate core business functions across different departments and processes within an organization.

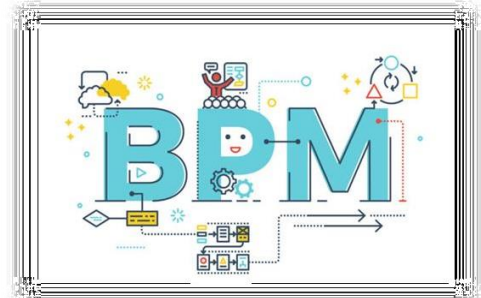
TOPIC 03

Business Process Management System Software



3.1 What?

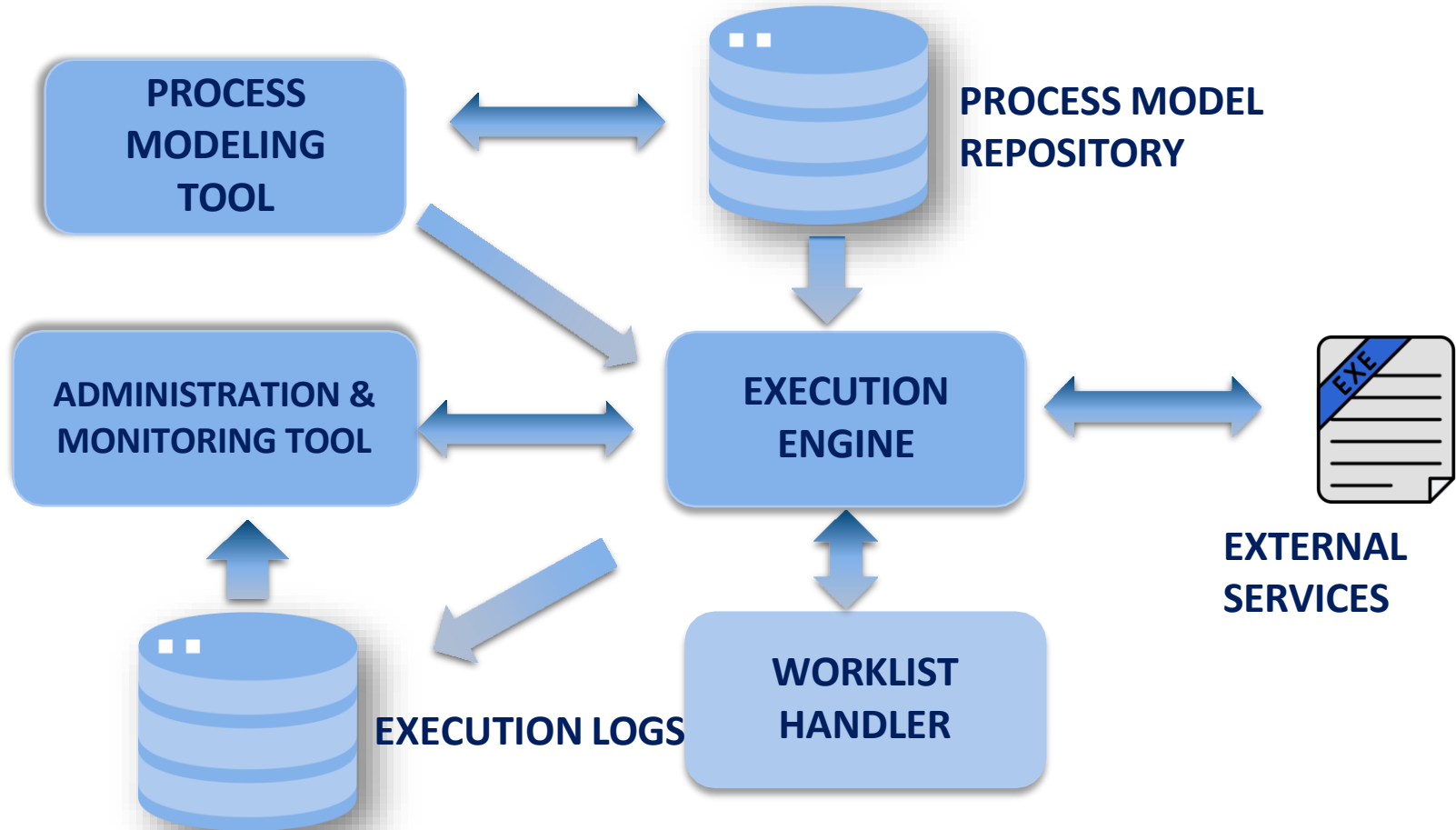
- Business Process Management Software (BPMS) refers to a type of **software platform** that enables organizations to **model**, **automate**, **manage**, **monitor**, and **optimize** their business processes.
- BPMS provides a comprehensive solution to **streamline** and **enhance** an organization's processes, making them more **efficient**, **effective**, and **adaptable to changes**.
- The purpose of a BPMS is to coordinate an **automated business process** in such a way that all work is done at the **right time** by the **right resource**.



3.2 Business Process Management System Software



3.3 Architecture of a BPMS



3.4 Architecture of a BPMS Explained..

1.Process Modeling Tool:

The Process Modeling Tool is a **graphical interface** that allows business analysts and process designers to **create**, **design**, and **model** business processes using visual representations such as **flowcharts**, **diagrams**, and **process maps**.

- To create and modify process models
- To annotate process models with additional data
- To store, share and retrieve process models from a **process model repository**

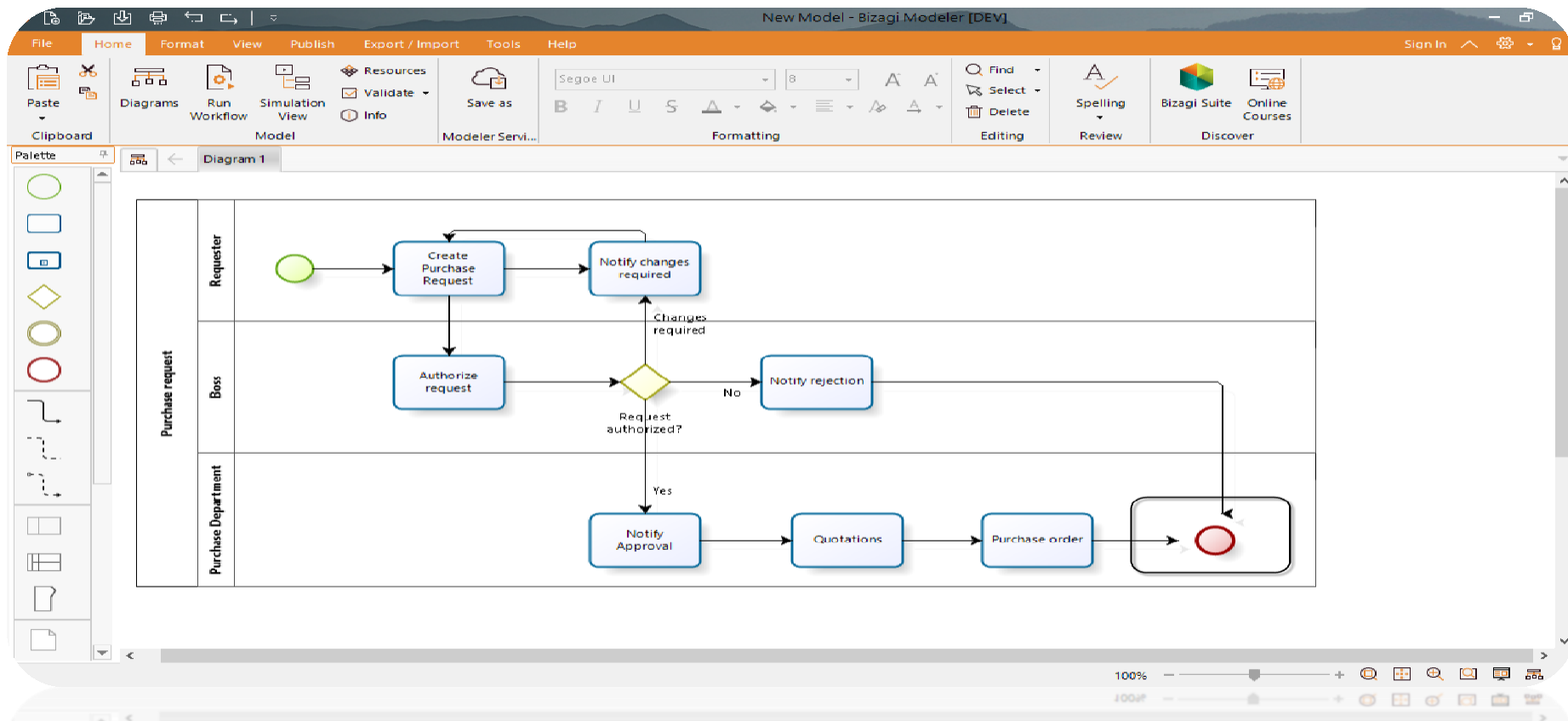
✓ This tool is mainly used during the design stage—not while the process is running.

Architecture of a BPMS Explained..

2.Execution Engine:

- The Execution Engine is the core component responsible for **executing** and **managing** automated business processes.
- It takes the process models created in the **Process Modeling Tool** and orchestrates the execution of **tasks**, **activities**, and **decisions** based on **predefined rules** and **logic**.
- The Execution Engine ensures that processes run smoothly and efficiently, routing tasks to the appropriate users or automated systems.
- Automatically **retrieves** and **store** data required for the execution

Process Modeler in Bizagi Suite



Architecture of a BPMS Explained..

3.Work-list Handler :

- Offers **work items** to process participants and allows participants to **commit** to these work items.
- Handles participants' **work queues** and **work item priorities**.
- May provide social network capabilities.
- This component ensures that users receive the **right tasks** at the **right time** and **facilitates collaboration** and **communication** among process participants.

Architecture of a BPMS Explained..

4.Administration & Monitoring Tools:

- Administration of all operational matters of a BPMS.
- To deal with **exceptional situations** (Eg : Remove outdated work items from the system.
- Monitor the performance of the **running business processes**.

Architecture of a BPMS Explained..

5.Process Repository & External Services:

- To involve **external applications** in the execution of a business process. (eg: rules engine, email or Twitter notification, DB connector, CRM connector)
- Whenever the execution engine needs a process model, it retrieves it from this repository.
- The BPMS also communicates with external services.
- These may include email systems, ERP systems, CRM systems, payment gateways, or databases.
- This allows different software applications to work together.

Work-list handler of Bizagi Suite

Bizagi Worklist Handler

The screenshot displays the Bizagi Worklist Handler web application. The browser address bar shows `www.localhost.com:1024/BuildIT/Default.htm`. The application has a navigation menu with links for Cases, Analysis Reports, Admin, Tools, and Log Out. The main content area is titled "App - BuildITModel - Extend the Renting time" and contains a form for managing equipment rental cases. The form includes fields for Reference nr, Cost per day, Requested start date, Expected end date, Comments to the supplier, Date of Delivery, and Issues found on delivery. There are also radio buttons for "Extend Renting Time" (Yes/No) and "Save" and "Next >>" buttons. A sidebar on the left shows "Bizagi Cases" with a sub-item "Expired (10)" and a list of "Smart Folders", "Bizagi Folders", "Bizagi Queries", and "Analysis Queries". The bottom status bar indicates the user is "Site Engineer (Europe/Helsinki)" on "Tuesday, May 22, 2012" at "1:30 PM".

Reference nr of the selected equipmet: 12345
Cost per day: \$65.00
Requested start date: Tuesday, May 22, 2012
Expected end date: Wednesday, May 09, 2012
Comments to the supplier: Must be with battery
Date of Delivery: Wednesday, May 30, 2012
Issues found on delivery:
Extend Renting Time: ☐ Yes ☐ No
Save Next >>
Creation Number: 153
Solution Date: 5/22/2012
Created by: Site Engineer
Current Assignee: Site Engineer
Current Event Assignees:
Please fill form fields.
Total Urgent Cases: 10
Site Engineer (Europe/Helsinki) Tuesday, May 22, 2012 1:30 PM

Monitoring Interface of ActiveMatrix BPM



Types of BPMS

- There are two ways to distinguish the available BPMSs.
 - The degree of support that the BPMS delivers
 - Their orientation on process or data
- Four types of BPMSs can be identified based on the above two specifications.
 - groupware systems
 - ad-hoc workflow systems
 - production workflow systems
 - case handling systems

3.4 Advantages of Introducing a BPMS

Workload Reduction

- Straight-through processing
- Less coordination
- Less gathering of relevant information

Flexible System Integration

- Easier to change process logic.
- Island automation

Execution Transparency

- Transparency of operational information
 - Transparency of historic information

Rule Enforcement

- Enforce separation of duties.
- Implement control tasks

3.5 Challenges in Introducing a BPMS

Technical Challenges

- Software have never previously built with the coordinated use in mind
- Connecting legacy systems
- Lack of process-awareness of traditional systems.
- Migrating existing process data to the new BPMS can be challenging

Organizational Challenges

- Balancing different stakeholder expectations
- Dynamic activities of organizations (e.g: Rule Changes)
- Perspective of users till they get hands on experience.
- Training employees to use the BPMS effectively and ensuring they understand its benefits can be challenging

TOPIC 04

Turning Process Models Executable



Turning Process Models into Executable Models

- Until now, we have created business process models mainly to help people understand how work is done.
- However, a BPMS cannot execute a simple business diagram directly.
- We first need to convert the business model into an executable process model that the BPMS can understand and execute automatically.

4.2 How?

A five-step method to incrementally transform a **business-oriented process model** into an executable one

1. Identifying automation boundaries
2. Reviewing manual tasks
3. Completing process model
4. Adjust task granularity
5. Specifying execution properties

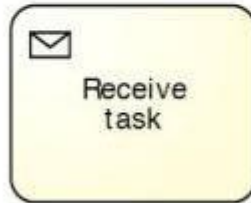
1. Identifying Automation Boundaries

Principle: **not all processes can be automated.**

- Start by identifying type of task
- Three types of tasks in a process
- **Automated Task** - performed by the BPMS itself or by an external service.
- **User Task** - performed by a participant with the assistance of the work-list handler of the BPMS or of an external task list manager.
- **Manual Tasks** – performed by process participants without the aid of any software.

1. Identifying Automation Boundaries

BPMN : Task Types

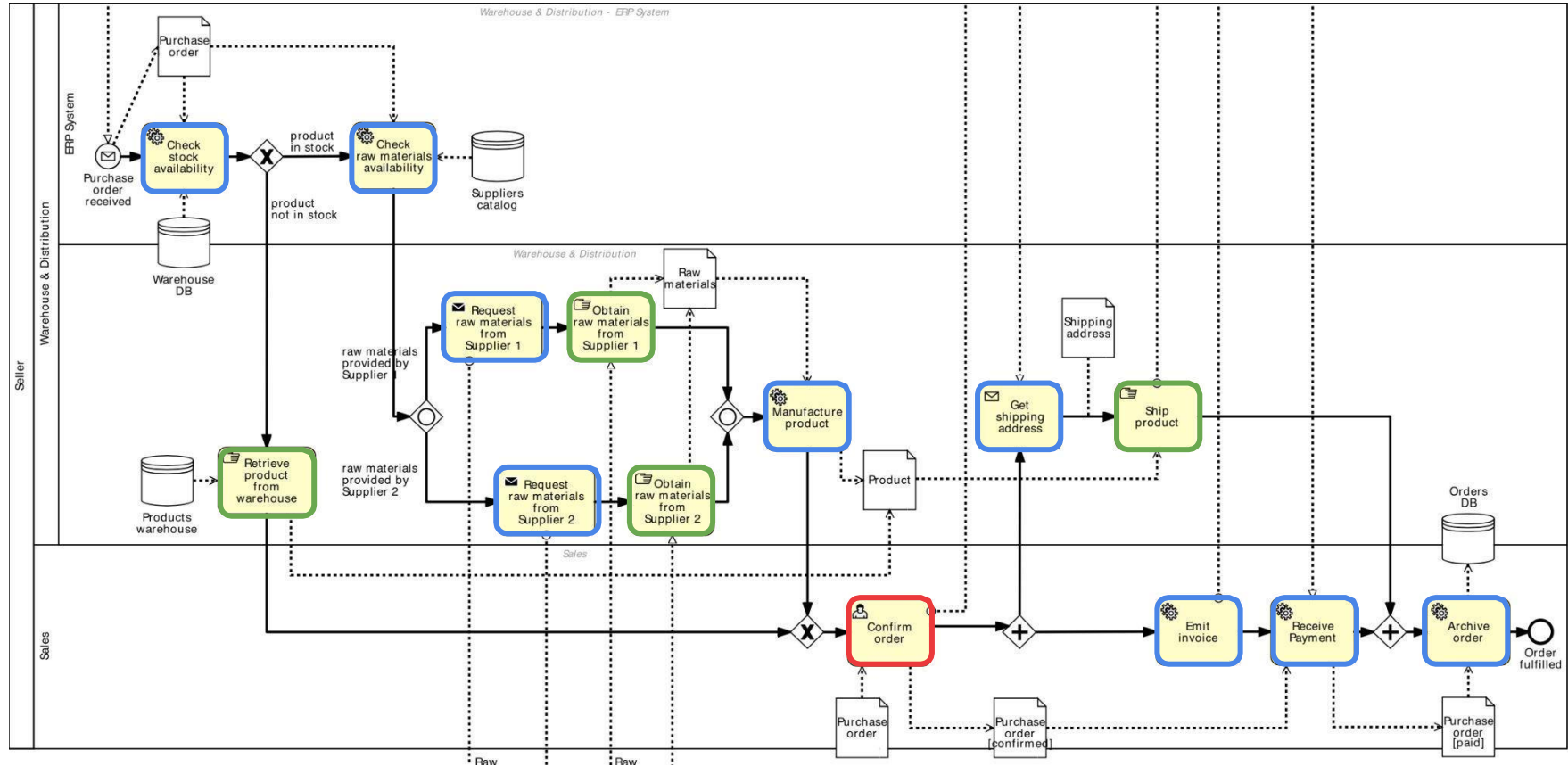


Automated tasks

User tasks

Manual tasks

Example :order fulfillment process model



So remember, the first step is not to automate everything immediately. Instead, we carefully identify which tasks are automated, which require users, and which remain manual. That helps us build an executable process model correctly.

2. Review Manual Tasks

Check whether the manual tasks can be linked to the BPMS

- Two ways of linking a manual task to a BPMS:
 - Implement it via a user task
 - Implement it via an automated task

2. Review Manual Tasks

1. Example : order fulfillment process model

A warehouse worker physically obtains the raw materials from the product warehouse.



 Obtain raw materials from Supplier 1



Using a barcode scanner to scan the raw materials and signal the completion of the task to the system.



 Obtain raw materials from Supplier 1

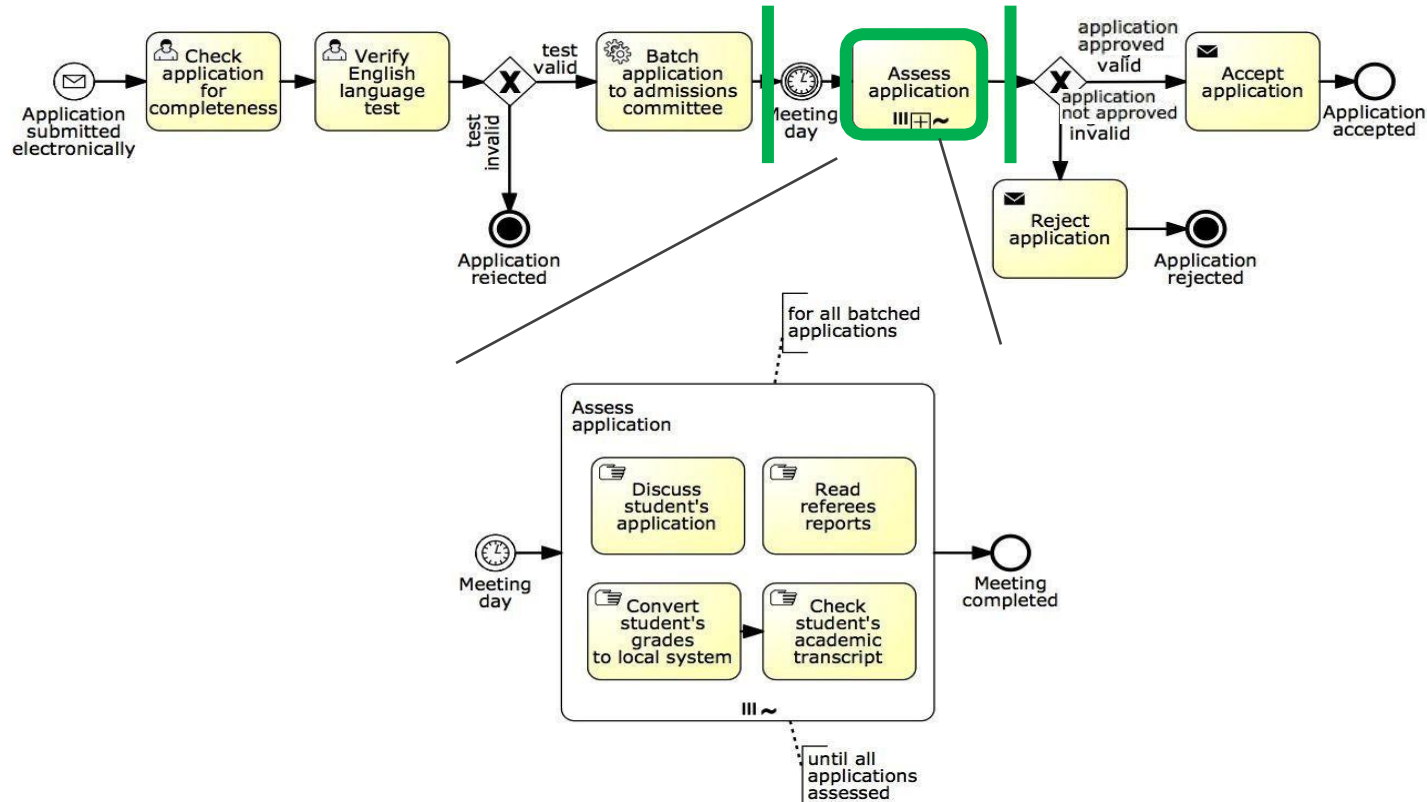
2. Review Manual Tasks

1. **Alternative**

- Isolate the manual task and focus on the automation of the process before and after it.

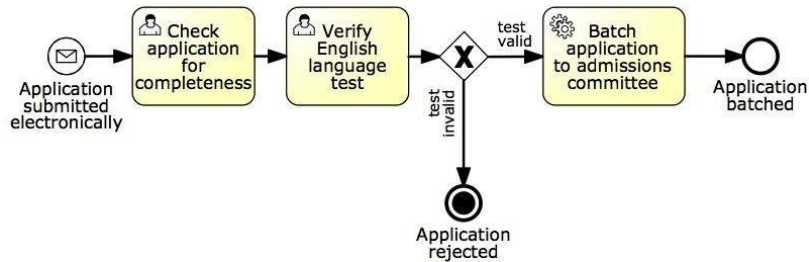
Alternative: Isolate manual tasks

Example : university admission process

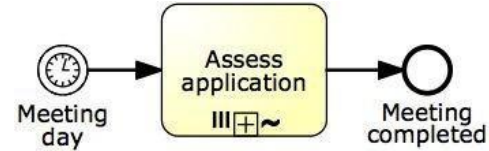


Alternative: Isolate manual tasks

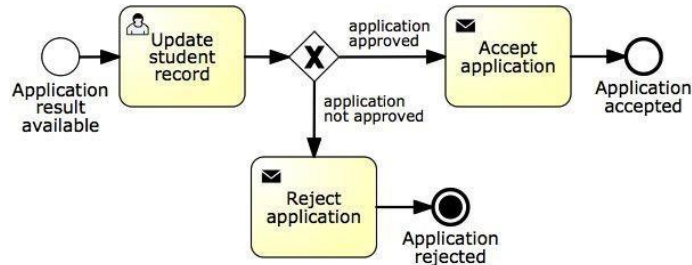
Example : university admission process



Segment 01



Segment 02



Segment 03

BPMN elements irrelevant for execution

- Physical data objects
 - Messages bearing physical data objects
 - Data stores (BPMS assumes that specific services with access to these data storage exist.)
 - Pools & lanes (used at a conceptual level only)
 - Text annotations
-
- Remove or neglect, depending on BPMS
 - **Example**

If we add a text note saying 'Manager should be careful when approving expensive orders,' it helps people reading the diagram, but the BPMS does not execute that note.

- **Key Message**

When converting a business model into an executable model, we keep only the information needed by the system.

3. Complete Process Model

Main goal: Check whether the process model is complete

- Two principles underlie this step:
 - i) **Exceptions are the rule**
 - This means unexpected situations happen frequently, so we must plan for them.
 - ii) **No data implies no decisions and no task handoff (so, better include how data is used and produced)**
 - This means every task must clearly define the data it receives and the data it produces.

3. Complete Process Model

i) Exceptions are the rule

Need to make sure that all exceptions are handled using appropriate exception handlers.

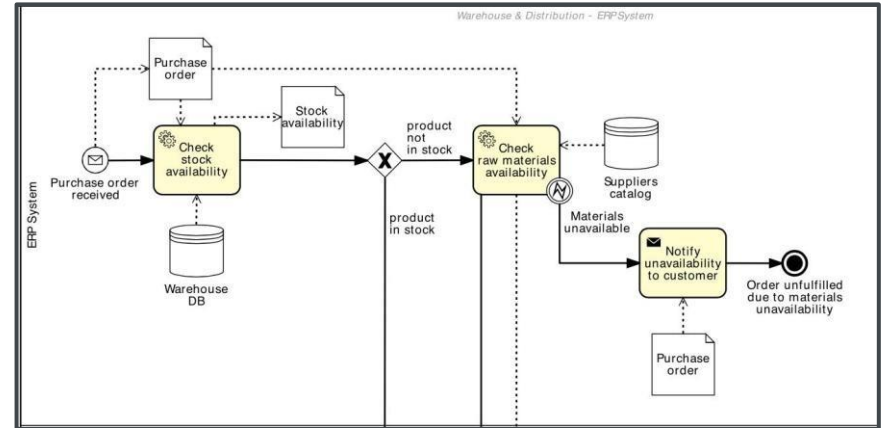
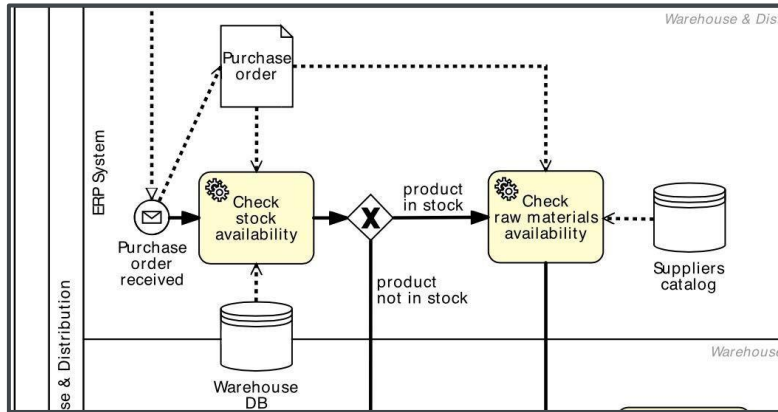
Example : The order fulfillment process may be aborted if;

- The materials required to manufacture the product are not available at the suppliers
- The customer sends an order cancellation.

3. Complete Process Model

Example :order fulfillment process model

Exception → The materials required to manufacture the product are not available at the suppliers



3. Complete Process Model

Example :order fulfillment process model

- Imagine the company wants to manufacture a product.
- Before production starts, the system checks whether the supplier has the required raw materials.
- If the materials are available, the process continues normally.
- If they are **not available**, the BPMS follows an alternative path.
- For example, it may:
 - Contact another supplier,
 - Wait until stock becomes available,
 - Or cancel the order.
- Instead of stopping with an error, the process follows a planned exception path.
- This is exactly why we include exception handling during process implementation

3. Complete Process Model

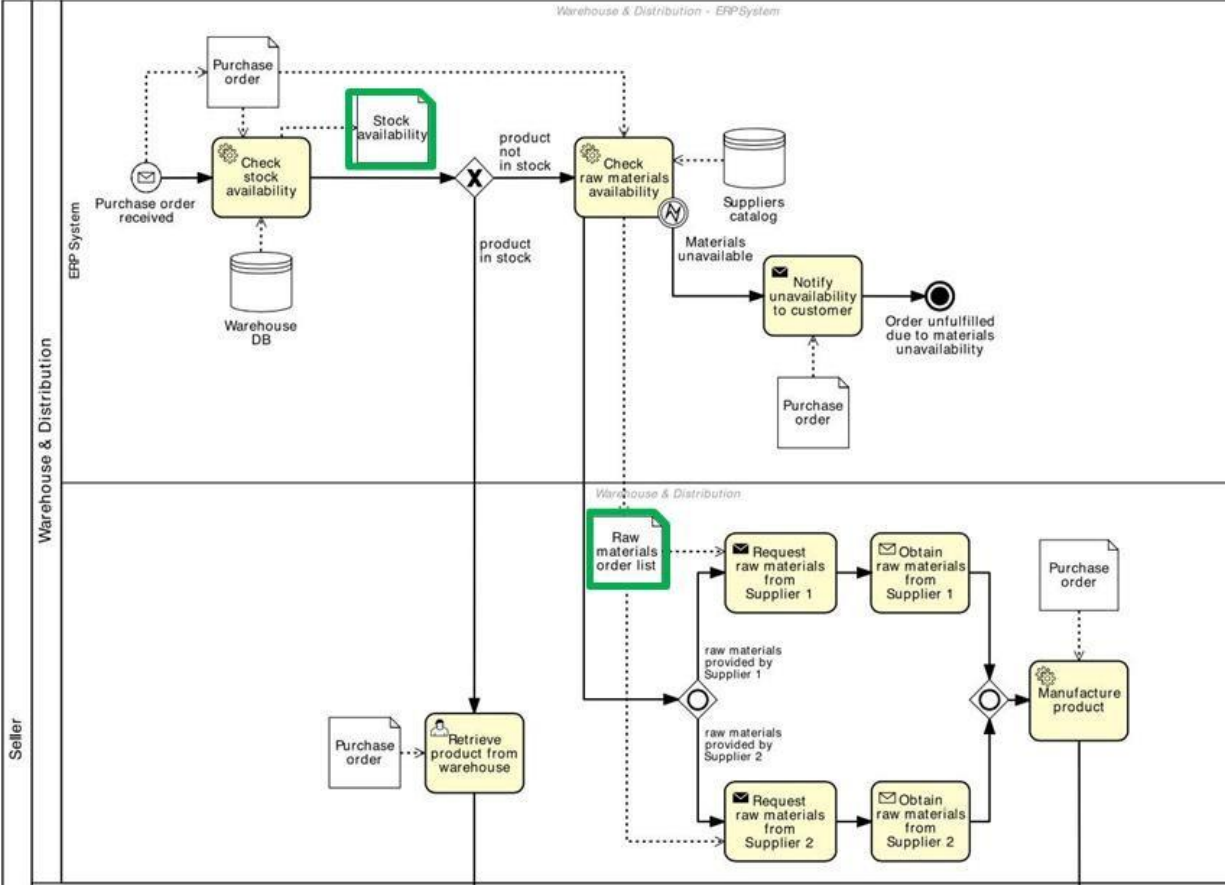
ii) No data implies no decisions and no task handoff

Need to specify all electronic data objects that are required as input and output by the tasks in the process.

Example :

- There is no input data object to task “Request raw materials from Supplier 1”, though this task does need the list of raw materials to be ordered.
- “Check stock availability” uses the Purchase order as input but does not produce any output data to store the results of the search.

Example : order fulfillment process model



4. Adjust Task Granularity

There is not necessarily a one-to-one mapping between the tasks in a business- oriented model and those in the corresponding executable model.

- Tasks in a conceptual model may not be at the right level of granularity for implementation. They may be either:
 - **too abstract**, in which case we need to decompose them
 - OR
 - **too detailed**, in which case they should be aggregated
- Principle: BPMSs add value if they coordinate handovers of work between resources.

4. Adjust Task Granularity

Name	Explanation	Prescribed action
Aggregation	Two or more consecutive tasks assigned to the same resource	Define a single user task

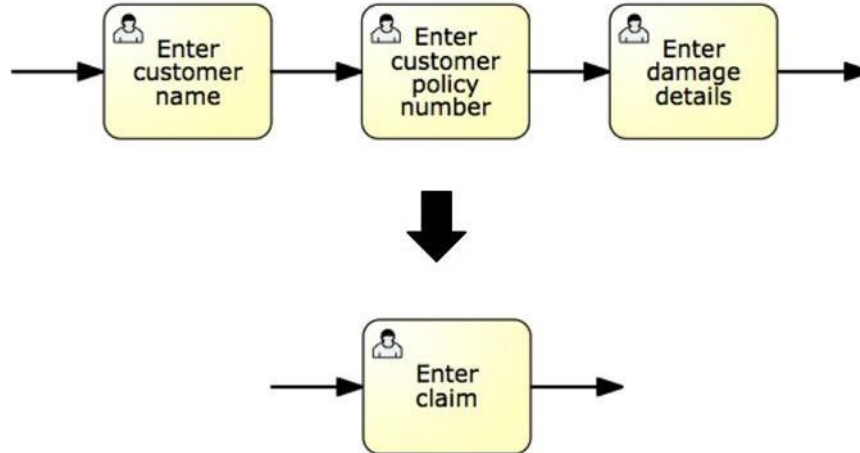
1. **Alternative**

- Sometimes it might be needed to keep consecutive tasks separate, even though they are performed by the same resource.
- These kind of cases could be handled by **adding an exception handler**.

4. Adjust Task Granularity

Example 01 :Aggregate any two consecutive tasks assigned to the same resource

Below sequence of user tasks should be aggregated into a single user task “Enter claim”, as all three tasks will be performed by the same claims handler.



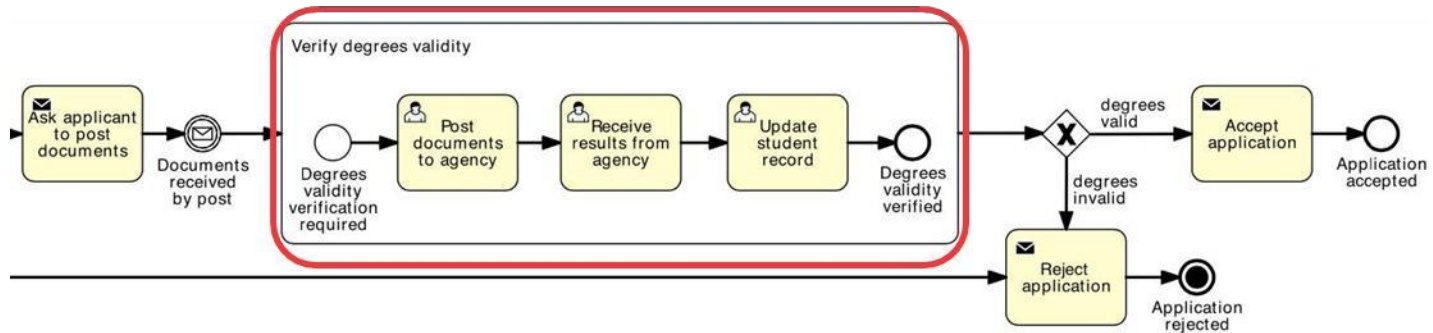
4. Adjust Task Granularity

Example 02 : adding an exception handler

Three user tasks within “**verify degree validity**” sub-process could be aggregated.

But it is required to monitor each task and manage potential exceptions

(Ex: results are not received within the given timeframe).



The best way to handle this delay is to add an exception handler to task “**Receive results from agency**”.

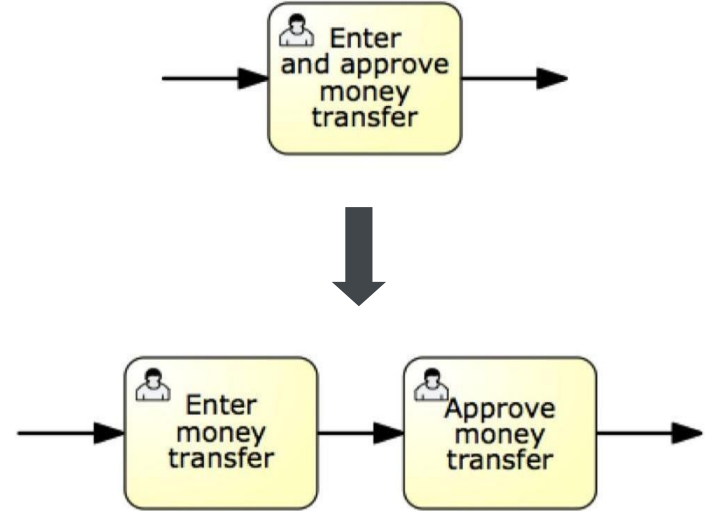
4. Adjust Task Granularity

Name	Explanation	Prescribed action
Coarse - Grained	Activity require more than one resource to be performed	Disaggregate it into more fine-grained tasks such that these can be assigned to different resource

Example : Refine tasks that are too coarse-grained

To enforce separation of duties, the following activity can be performed by **two different participants**, even if they have the same role.

- First, a financial officer enters the order, then a different financial officer approves it.



5. Specify Execution Properties

Specify how each model element is effectively implemented by the BPMS.

Example : “Check stock availability”

Saying this task requires the **purchase order** as input to contact the warehouse ERP system is not enough. We need to specify;

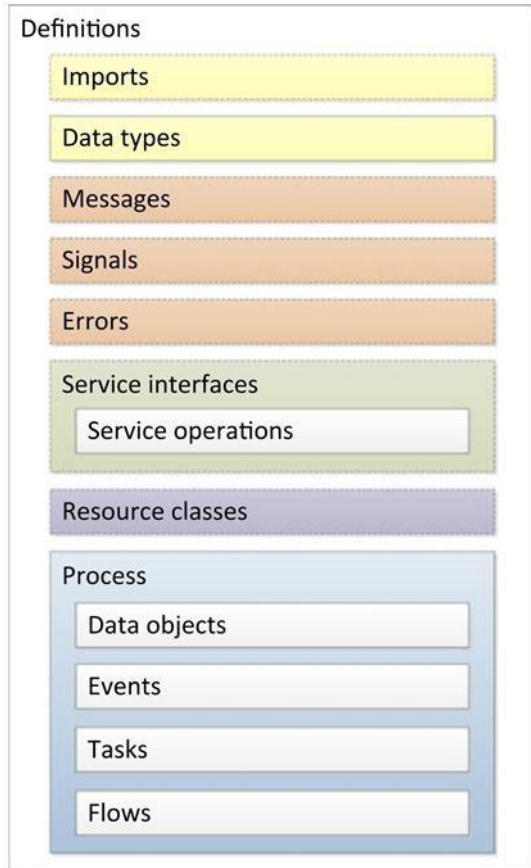
- The service provided by the ERP system to check stock levels
 - Its location in the network
 - The product information in the purchase order (The format of the input object)
 - The information produced by the service (The format of the output object).
- These implementation details are called execution properties.

5. Specify Execution Properties

Execution properties

- Process variables, messages, signals and errors
- Task and event variables and their mappings to process variables
- Service details
- Code snippets
- Participant assignment rules and user interface structure for user tasks
- Task, event and sequence flow expressions
- BPMS-specific properties

Structure of the BPMN format



```
<purchaseOrder>
  <order>
    <orderNumber>15664</orderNumber>
    <orderDate>2012-10-23</orderDate>
    <status>confirmed</status>
    <currency>EUR</currency>
    <productCode>345-EAR</productCode>
    <quantity>10</quantity>
  </order>
  <customer>
    <name>John</name>
    <surname>Brown</surname>
    <address>
      <street>8 George St</street>
      <city>Brisbane</city>
      <state>Queensland</state>
      <postCode>4000</postCode>
      <country>Australia</country>
    </address>
    <phone>+61 7 3240 0010</phone>
    <fax>+61 7 3221 0412</fax>
  </customer>
</purchaseOrder>
```

XML representation of a particular purchase order instance



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LESSON 08

Recap...

Thank You!